

**Table 1: Verster et al. (2021) - Predicting Home-Loan Take-Up**

|                                 | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Title</b>                    | <b>Predicting take-up of home loan offers using tree-based ensemble models: A South African case study.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Harvard reference</b>        | Verster, T., Harcharan, S., Bezuidenhout, L. and Baesens, B. (2021) 'Predicting take-up of home loan offers using tree-based ensemble models: A South African case study', <i>South African Journal of Science</i> , 117(1/2), Art. #7607. Available at: <a href="https://doi.org/10.17159/sajs.2021/7607">https://doi.org/10.17159/sajs.2021/7607</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Link</b>                     | <a href="https://doi.org/10.17159/sajs.2021/7607">https://doi.org/10.17159/sajs.2021/7607</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Modelling</b>                | <p>The study modelled home loan take-up as a binary classification problem i.e where customers either accepted or rejected the loan offer.</p> <p>Logistic regression was used as the baseline model and compared with tree-based ensemble methods (bagging and boosting).</p> <p>Additional techniques, including neural networks and support vector machines, were evaluated but performed less effectively than bagging and boosting.</p> <p>Bagging and boosting achieved validation Gini coefficients of 0.467 and 0.469, outperforming logistic regression (0.403).</p> <p>The results demonstrated that more complex ensemble models achieved higher predictive accuracy than logistic regression, but this improvement came at the cost of reduced interpretability. This highlights the importance of balancing model performance with the ability to explain predictions, particularly in business environments where model outputs must support decision-making.</p>                                                                                                                                |
| <b>Preprocessing techniques</b> | <p>The study undertook a structured data preparation process before model development. The dataset was first divided into 70% training and 30% validation sets using stratified sampling to ensure that the proportions of take-up and non-take-up customers were maintained across both datasets. This allowed model performance to be evaluated on unseen data.</p> <p>Data transformation techniques were then applied to improve the suitability of the data for modelling. Interest rates were adjusted relative to the South African repo rate to account for changes in the broader economic environment, categorical variables were converted into indicator variables, and selected numerical variables were scaled.</p> <p>Feature selection formed part of the preprocessing stage. Stepwise regression was used to identify the variables that contributed most significantly to predicting home loan take-up, resulting in six variables being retained for modelling. The prepared dataset was subsequently used to train and validate the logistic regression, bagging and boosting models.</p> |

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| <b>Datasets used</b> | The study used 294,479 home loan approvals from a South African bank between January 2010 and July 2015. Variables included demographic information, customer risk grades, employment type, loan characteristics, interest rates, loan-to-value ratios (LTV), loan type and indicators showing whether customers were new to the bank's home-loan products.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Key takeaways</b> | <p>Customer conversion can be modelled successfully using historical data. The study demonstrates that customer take-up behaviour can be predicted using demographic, behavioural and product-related variables, which is directly relevant to predicting first-deposit activation at STADIOEquities.</p> <p>Model comparison is important. While logistic regression provided an interpretable baseline, ensemble models achieved superior predictive performance.</p> <p>Data preparation materially influences model performance. The use of feature selection, variable transformation and robust validation improved predictive accuracy. Similar preprocessing techniques should be applied to customer, onboarding, behavioural, marketing and support data before modelling first-deposit activation.</p> |

**Table 2: Breed and Verster (2017) - Predicting Customer Response**

| Field                    | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Title</b>             | <b>The benefits of segmentation: Evidence from a South African bank and other studies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Harvard reference</b> | Breed, D.G. and Verster, T. (2017) 'The benefits of segmentation: Evidence from a South African bank and other studies', <i>South African Journal of Science</i> , 113(9/10), Article 2016-0345, pp. 1–7. doi:10.17159/sajs.2017/20160345.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Link</b>              | <a href="https://doi.org/10.17159/sajs.2017/20160345">https://doi.org/10.17159/sajs.2017/20160345</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Modelling</b>         | <p>The study modelled customer response as a binary classification problem, where customers either responded or did not respond to a direct-marketing campaign.</p> <p>A range of modelling approaches was evaluated, including logistic regression, segmentation-based logistic regression, neural networks, support vector machines (SVMs), memory-based reasoning, decision trees and gradient boosting. Stepwise selection was used for logistic regression to identify the most relevant predictors.</p> <p>Model performance was assessed using the validation Gini coefficient. For the South African banking dataset. Gradient boosting achieved the highest</p> |

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|                                 | <p>validation Gini coefficient (35.31%), followed by supervised segmentation combined with logistic regression (33.70%).</p> <p>The findings suggest that both advanced machine learning techniques and customer segmentation can improve predictive performance when modelling customer behaviour.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Preprocessing techniques</b> | <p>The study followed a structured data preparation process before model development.</p> <p>All explanatory variables were standardised, placing variables on a comparable scale.</p> <p>The data were then randomly divided into development (training) and validation sets, with 50% allocated to each, allowing model performance to be evaluated on unseen data.</p> <p>Additional preprocessing techniques such as the treatment of missing values and dataset balancing for rare target events were applied.</p> <p>For the South African direct marketing dataset specifically, the reported preprocessing step was standardisation of the explanatory variables.</p>                                                                                                                                                                                                                                                                                                                                       |
| <b>Datasets used</b>            | <p>The study evaluated the modelling techniques across six datasets.</p> <p>The most relevant dataset was a South African banking direct marketing dataset containing customer information, product holdings, utilisation measures and behavioural variables. The target variable indicated whether a customer responded to a personal-loan marketing campaign.</p> <p>The dataset contained 4,720 observations and 24 explanatory variables describing the customers, the banking products they held, and their utilisation and behaviour regarding those products. The binary target variable indicated whether each customer responded to a direct marketing campaign for a personal loan or not.</p> <p>Five additional datasets, sourced from the UCI Machine Learning Repository and Kaggle, covered credit applications, insurance claims, wine quality, protein structure and chess outcomes. They were used to compare the performance of the modelling techniques across different application areas.</p> |
| <b>Key takeaways</b>            | <p>Customer segmentation can improve predictive performance. The study demonstrated that separating customers into more homogeneous groups can enhance a model's ability to distinguish between different customer outcomes. This is relevant to STADIOEquities because activation behaviour may differ across customer segments and onboarding journeys.</p> <p>Model selection should be evidence-based. No single modelling technique performed best across all datasets. This supports evaluating multiple candidate models on the STADIOEquities dataset rather than selecting an algorithm before exploratory analysis and model testing.</p>                                                                                                                                                                                                                                                                                                                                                                 |

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|  | Data preparation and validation are critical components of the modelling process. The study standardised variables and evaluated model performance on an independent validation dataset. A similar approach should be followed when analysing first-deposit activation to ensure that the findings generalise beyond the training data and provide reliable business insights. |
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**Table 3: Moro, Cortez and Rita (2014) - Predicting Bank Telemarketing Success**

| Field                           | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Title</b>                    | <b>A data-driven approach to predict the success of bank telemarketing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Harvard reference</b>        | Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', <i>Decision Support Systems</i> , 62, pp. 22–31. doi:10.1016/j.dss.2014.03.001.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Link</b>                     | <a href="https://repositorio.biblioteca.iscte-iul.pt/bitstream/10071/9499/5/dss_v3.pdf">https://repositorio.biblioteca.iscte-iul.pt/bitstream/10071/9499/5/dss_v3.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Modelling</b>                | The study predicted used a binary classification problem to evaluate whether a customer would subscribe to a long-term deposit following a telemarketing campaign. Four modelling techniques were compared: logistic regression, decision trees, neural networks and SVMs. A semi-automatic feature selection process, incorporating business knowledge and a forward selection method, reduced the original 150 variables to 22 features for modelling. Model performance was evaluated using AUC (Area Under the Curve) and ALIFT (Average Lift), together with a realistic rolling-window validation approach that tested model performance on more recent customer data. The neural network achieved the strongest reported results, with an AUC of 0.80 and an ALIFT of 0.70. |
| <b>Preprocessing techniques</b> | A structured data preparation process was implemented before model development. Customer, product and socio-economic variables were explored, resulting in approximately 150 candidate variables. A semi-automatic feature selection process was then applied to identify the 22 variables most relevant to predicting customer response. The researchers also implemented a rolling-window evaluation approach, where older observations were used for model development and newer observations were used for model testing. This enabled the models to be evaluated on future customer behaviour rather than solely on historical observations.                                                                                                                                  |
| <b>Datasets used</b>            | The study used a real-world dataset from a Portuguese retail bank containing 52,944 customer records collected between May 2008 and June 2013. The data included customer information, product information and economic indicators related to telemarketing campaigns. The outcome measured whether a customer subscribed to a long-term deposit after being contacted by the bank.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Key takeaways</b>            | Feature selection improves model development. The study reduced 150 potential variables to 22 relevant features, demonstrating the importance of                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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|  | <p>identifying variables that contribute meaningful predictive information before modelling.</p> <p>Multiple modelling approaches should be evaluated. Logistic regression, decision trees, neural networks and SVMs produced different levels of performance. This highlights the importance of comparing several candidate models before selecting a final approach.</p> <p>Validation should reflect real world use. The rolling-window approach tested model performance on newer customer data, making the evaluation more realistic. This is relevant to STADIOEquities because the objective is to use historical customer behaviour to identify customers who are likely to activate in the future. because the objective is to use historical customer information to identify customers who are likely to activate in the future.</p> |
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**Table 4: Selected Proxy Dataset - Bank Marketing Dataset**

| Field                                    | Summary                                                                                                                                                                                                                                                         |
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| <b>Name</b>                              | Bank Marketing Dataset                                                                                                                                                                                                                                          |
| <b>The title of the dataset</b>          | Bank Marketing                                                                                                                                                                                                                                                  |
| <b>Harvard reference</b>                 | Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', Decision Support Systems, 62, pp. 22–31. doi:10.1016/j.dss.2014.03.001.                                                                         |
| <b>Link</b>                              | <a href="https://archive.ics.uci.edu/dataset/222/bank+marketing">https://archive.ics.uci.edu/dataset/222/bank+marketing</a>                                                                                                                                     |
| <b>Number of instances</b>               | The dataset contains 45,211 customer records in the bank-full.csv version.                                                                                                                                                                                      |
| <b>Number of feature columns</b>         | 16 input features, excluding the target variable. These include customer demographic, financial and campaign-related variables.                                                                                                                                 |
| <b>Number of classes (if applicable)</b> | 2 classes: yes and no, indicating whether a customer subscribed to a term deposit.                                                                                                                                                                              |
| <b>Target range (if applicable)</b>      | Not applicable, as this is a classification problem rather than a regression problem.                                                                                                                                                                           |
| <b>Origin</b>                            | The dataset originates from direct marketing campaigns conducted by a Portuguese banking institution. Customer information, campaign interactions and financial characteristics were collected to predict whether a customer would subscribe to a term deposit. |

